

PART-A

MODEL ANSWERS OF END-SEMESTER EXAMINATION (TOTAL POINTS: 20)

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$\begin{aligned} Y_1 &= \theta_1 + \theta_2 + \epsilon_1, \\ Y_2 &= \theta_1 - 2\theta_2 + \epsilon_2, \\ Y_3 &= 2\theta_1 - \theta_2 + \epsilon_3, \end{aligned}$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

Solution: We have

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}_{3 \times 1} = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}_{3 \times 2} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}_{2 \times 1} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{pmatrix} \Rightarrow Y = X\theta + \epsilon.$$

The least square estimator of θ is $\hat{\theta} = (X'X)^{-1}X'Y$. Now,

$$\begin{aligned} (X'X) &= \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}_{3 \times 2} \\ &= \begin{bmatrix} 1+1+4=6 & 1-2-2=-3 \\ 1-2-2=-3 & 1+4+1=6 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}. \\ (X'X)^{-1} &= \frac{1}{36-9} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix} = \frac{3}{27} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \end{aligned}$$

Thus

$$\begin{aligned} \hat{\theta} &= \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}_{2 \times 3} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} y_1 + y_2 + 2y_3 \\ y_1 - 2y_2 - y_3 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 2(y_1 + y_2 + 2y_3) + (y_1 - 2y_2 - y_3) \\ (y_1 + y_2 + 2y_3) + 2(y_1 - 2y_2 - y_3) \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 3y_1 + 3y_3 \\ 3y_1 - 3y_2 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{3}(y_1 + y_3) \\ \frac{1}{3}(y_1 - y_2) \end{bmatrix}. \end{aligned}$$

Hence, $\hat{\theta}_1 = \frac{1}{3}(y_1 + y_3)$ and $\hat{\theta}_2 = \frac{1}{3}(y_1 - y_2)$.

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

- (a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

Solution:

To compute correlation matrix ρ of X .

$$E(U_1) = E(U_2) = \frac{1}{2} \quad \text{and} \quad V(U_1) = V(U_2) = \frac{1}{12}.$$

So, $V(X_1) = V(X_2) = \frac{1}{12}$ and $V(X_3) = V(U_1) + V(U_2) = \frac{2}{12}$.

$$\begin{aligned} \text{Cov}(X_1, X_2) &= \text{Cov}(U_1, U_2) = 0 = \text{Cov}(X_2, X_1) \\ \text{Cov}(X_1, X_3) &= \text{Cov}(U_1, U_1 + U_2) \\ &= \text{Cov}(U_1, U_1) \\ &= V(U_1) = \frac{1}{12} = \text{Cov}(X_3, X_1) \\ \text{Cov}(X_2, X_3) &= \text{Cov}(U_2, U_1 + U_2) \\ &= V(U_2) = \frac{1}{12} = \text{Cov}(X_3, X_2). \end{aligned}$$

Now, $\text{Corr}(X_i, X_i) = 1$ for $i = 1, 2, 3$. Moreover,

$$\begin{aligned} \text{Corr}(X_1, X_2) &= 0, \\ \text{Corr}(X_1, X_3) &= \text{Corr}(X_3, X_1) = \frac{\text{Cov}(X_1, X_3)}{\sqrt{V(X_1)V(X_3)}} = \frac{\frac{1}{12}}{\sqrt{\frac{1}{12} \cdot \frac{2}{12}}} = \frac{1}{\sqrt{2}}, \\ \text{Corr}(X_2, X_3) &= \frac{\text{Cov}(X_2, X_3)}{\sqrt{V(X_2)V(X_3)}} = \frac{1}{\sqrt{2}}. \end{aligned}$$

Hence, the correlation matrix ρ of X is

$$\rho = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}.$$

- (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

Solution: Let Y_1 be the first principal component based on the correlation matrix ρ . Then

$$V(Y_1) = \lambda_1,$$

where λ_1 is the highest eigenvalue of the matrix ρ . The eigenvalues of ρ are solution of the following equation:

$$\begin{aligned} (1-\lambda) \begin{vmatrix} 1-\lambda & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 1-\lambda \end{vmatrix} - 0 \begin{vmatrix} 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 1-\lambda \end{vmatrix} + \frac{1}{\sqrt{2}} \begin{vmatrix} 0 & 1-\lambda \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{vmatrix} &= 0 \\ \Rightarrow (1-\lambda) \left[(1-\lambda)^2 - \frac{1}{2} \right] - 0 + \frac{1}{\sqrt{2}} \left[0 - \frac{1}{\sqrt{2}}(1-\lambda) \right] &= 0 \\ \Rightarrow (1-\lambda)^3 - \frac{1}{2}(1-\lambda) - \frac{1}{2}(1-\lambda) &= 0 \\ \Rightarrow (1-\lambda)^3 - (1-\lambda) &= 0 \\ \Rightarrow (1-\lambda)((1-\lambda)^2 - 1) &= 0 \\ \Rightarrow (1-\lambda)(\lambda^2 - 2\lambda) &= 0 \\ \Rightarrow \lambda(1-\lambda)(\lambda-2) &= 0. \end{aligned}$$

Hence, the largest eigenvalue is 2. So,

$$V(Y_1) = 2.$$

3. Let the matrix of distance be

$$\begin{array}{c} \begin{array}{cccc} & 1 & 2 & 3 & 4 \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 \\ 1 \\ 11 \\ 5 \end{bmatrix} & \begin{bmatrix} \\ 0 \\ 2 \\ 3 \end{bmatrix} & \begin{bmatrix} \\ \\ 0 \\ 3.5 \end{bmatrix} & \begin{bmatrix} \\ \\ \\ 0 \end{bmatrix} \end{array}$$

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

Solution: Complete linkage: At the first stage merge objects 1 and 2 together as they are most similar.

$$\begin{array}{c} \begin{array}{cccc} & 1 & 2 & 3 & 4 \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 \\ \textcircled{1} \\ 11 \\ 5 \end{bmatrix} & \begin{bmatrix} \\ 0 \\ 2 \\ 3 \end{bmatrix} & \begin{bmatrix} \\ \\ 0 \\ 3.5 \end{bmatrix} & \begin{bmatrix} \\ \\ \\ 0 \end{bmatrix} \end{array}$$

At stage 2, we compute distances using complete linkage:

$$\begin{aligned} d_{(12),3} &= \max(d_{13}, d_{23}) = \max(11, 2) = 11, \\ d_{(12),4} &= \max(d_{14}, d_{24}) = \max(5, 3) = 5. \end{aligned}$$

So, we obtain the distance matrix,

$$\begin{array}{c} \begin{array}{ccc} & (12) & 3 & 4 \\ \begin{array}{c} (12) \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 \\ 11 \\ 5 \end{bmatrix} & \begin{bmatrix} \\ 0 \\ \textcircled{3.5} \end{bmatrix} & \begin{bmatrix} \\ \\ 0 \end{bmatrix} \end{array}$$

At stage 3, merge objects 3 & 4, as they are most similar.

Now, we compute,

$$\begin{aligned} d_{(12)(34)} &= \max\{d_{(12)3}, d_{(12)4}\} \\ &= \max\{11, 5\} \\ &= 11 \end{aligned}$$

So, we obtain the new distance matrix,

$$\begin{array}{c} \begin{array}{cc} & (12) & (34) \\ \begin{array}{c} (12) \\ (34) \end{array} & \begin{bmatrix} 0 \\ \textcircled{11} \end{bmatrix} & \begin{bmatrix} \\ 0 \end{bmatrix} \end{array}$$

Now, at the final stage, merge clusters (12) & (34) together to form a single cluster.

Average linkage: At the first stage merge objects 1 and 2 together as they are most similar. At stage 2, we compute distances using complete linkage:

$$d_{(12),3} = \frac{d_{13} + d_{23}}{2} = \frac{11 + 2}{2} = 6.5,$$

$$d_{(12),4} = \frac{d_{14} + d_{24}}{2} = \frac{5 + 3}{2} = 4.$$

So, we obtain the distance matrix,

$$\begin{array}{c} (12) \\ 3 \\ 4 \end{array} \begin{bmatrix} & (12) & 3 & 4 \\ \begin{bmatrix} 0 & & \\ 6.5 & 0 & \\ 4 & \textcircled{3.5} & 0 \end{bmatrix} \end{bmatrix}$$

At stage 3, merge objects 3 & 4, as they are most similar. Now, we compute,

$$d_{(12)(34)} = \frac{d_{(12)3} + d_{(12)4}}{2} = \frac{6.5 + 4}{2} = 5.25.$$

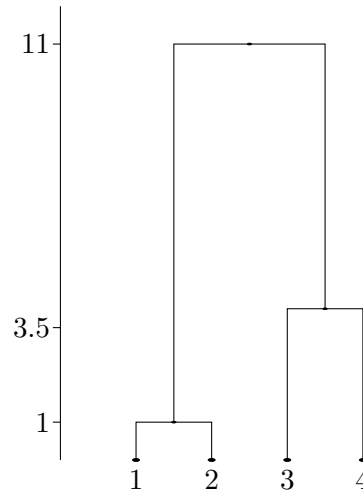
So, we obtain the new distance matrix,

$$\begin{array}{c} (12) \\ (34) \end{array} \begin{bmatrix} & (12) & (34) \\ \begin{bmatrix} 0 & \\ \textcircled{5.25} & 0 \end{bmatrix} \end{bmatrix}$$

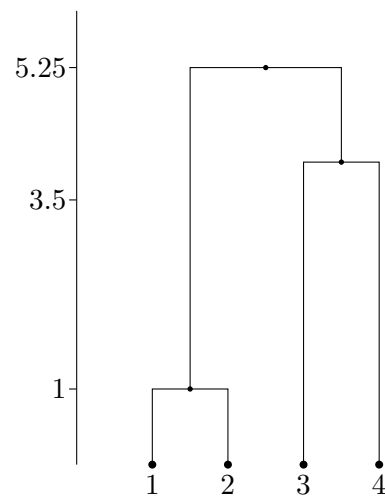
Now, at the final stage, merge clusters (12) & (34) together to form a single cluster.

- (b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

Solution: Dendrogram for complete linkage:



Dendrogram for average linkage:



Comparing the two dendrograms, we observe that the single linkage and complete linkage do not differ in allocating objects in this case. However, the distance at which different sets are merged are different in these two procedure.